Fixed-charge bilocals and selected preparations

Two-component memory, strong reconstruction and joint moments

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Abstract

We connect the fixed-charge ordinary and neutral field estimates to the actual centered bilocal. The charged OU identity and an even auxiliary-field expansion first give all fixed finite Schatten bounds on the total memory. At half filling, the central Fourier source theorem extends to colored words, proving the required four-field conditional regression. The two memory components then have their common local limit. A static-memory dilation gives an $O(p^{-1})$ strong reconstruction of the actual bilocal from actual memory. Consequently finite products of covariant bilocals converge between the already controlled fixed polynomial selected preparations.

1 Model, objects and statements

Use the common static Gaussian overlapping p-body CAR model of [1], standard CAR, $\hbar = 1$, fixed $J_s > 0$ and string time. Let $\lambda_N = p^2/N \to \lambda > 0$, $h = H_m$, $d_m = \binom{N}{m}$, $C_m I = \mathbb{E}h^2$, and $\|X\|_{r,m} = (\mathbb{E}d_m^{-1} \operatorname{Tr}|X|^r)^{1/r}$. All norm orders $1 \le r < \infty$ and real-time windows are fixed. Uniform bounds are on $|m - N/2| \le L_0 p$ with fixed L_0; limiting moments below are at even N and $m = N/2$.

Keep the exact dilation from [1]:

$$V\psi = m^{-1/2} \sum_i |i\rangle c_i \psi, \quad K = I_N \otimes H_{m-1}, \quad Q = I - VV^\dagger,$$

$$a = V^\dagger K V = \alpha h, \quad \alpha = 1 - p/m, \quad R = QKV, \quad D_\perp = QKQ, \quad G = R^\dagger R,$$

$$F(t) = V^\dagger e^{itK} V, \quad M(s) = R^\dagger e^{isD_\perp} R, \quad \Lambda(s) = R^\dagger e^{isK} R.$$

Let $\mathcal{P}_h X = \sum_e P_e X P_e$ retain whole degenerate energy blocks. The actual fields are

$$\mathcal{F}_N(s) = (1 - \mathcal{P}_h) m e^{-ihs/2} M(s) e^{-ihs/2}, \qquad \mathcal{B}_N(t) = (1 - \mathcal{P}_h) m e^{-iht/2} F(t) e^{-iht/2}.$$

Theorem 1. *The fields $\mathcal{F}_N, \mathcal{B}_N$ have uniformly bounded fixed finite norms on the stated charge and time windows. For $t \ge 0$,*

$$\left\| \mathcal{B}_N(t) + \int_0^t ds \int_{-(t-s)/2}^{(t-s)/2} du\, e^{ihu} \mathcal{F}_N(s) e^{-ihu} \right\|_{r,m} = O_{r,S,L_0}(p^{-1}). \tag{1}$$

At half filling, their finite ordered covariant moments, including blocks between the fixed polynomial preparations specified in Section 8, converge to the common local fields in Section 6. Negative times use adjoints.

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2 The charged identity with finite alpha retained

Define $\mathcal{O}_\alpha = \alpha^{\mathcal{N}}$ on coupling Gaussian chaos, $\epsilon = 1 - \alpha^2$, and the body loss on full charge families

$$(DX)_m = mX_m - \sum_i c_i^\dagger X_{m-1} c_i.$$

Introduce string-time fields

$$\begin{aligned} E_m(s) &= m[F(s) - \mathcal{O}_\alpha(e^{ish})] = [m(1-\alpha^{\mathcal{N}}) - D]e^{isH}\big|_m, \\ \mathsf{G}_m(s) &= [p\mathcal{N} - D - \lambda_N \mathcal{N}(\mathcal{N}-1)]e^{isH}\big|_m, \\ \mathscr{Q}_\alpha[X] &= -X'' + i\alpha\{h, X'\} + \alpha^2 hXh. \end{aligned}$$

Derivatives act on s, not on disorder. Direct differentiation of $F = V^\dagger e^{isK} V$ gives $\Lambda = \mathscr{Q}_\alpha[F]$. For an independent copy u of h, Gaussian OU averaging gives

$$\Lambda_{\mathrm{OU}}(s) := \mathscr{Q}_\alpha[\mathcal{O}_\alpha(e^{ish})] = \epsilon \mathbb{E}_u[u e^{is(\alpha h + \sqrt{\epsilon}u)} u], \quad m(\Lambda - \Lambda_{\mathrm{OU}}) = \mathscr{Q}_\alpha[E_m]. \tag{2}$$

To verify the first equality, set $v = \alpha h + \sqrt{\epsilon}u$; v commutes with its own exponential. Expanding $(v - \alpha h)e^{isv}(v - \alpha h)$ gives exactly the derivative expression. The two outer u's are the same auxiliary copy as the one inside the exponential. These identities are inherited from [7, 8], with the factors of m displayed.

The difference $E_m - \mathsf{G}_m$ is the scalar chaos multiplier $r_m(\mathcal{N})e^{ish}$, where

$$\begin{aligned} r_m(k) &= m(1-\alpha^k) - pk + \lambda_N k(k-1), \\ |r_m(k)| &\le a_m (k+1)^3, \quad a_m = \frac{4\lambda_+ |m - N/2|}{N} + \frac{8\lambda_+^2}{3p} = O(p^{-1}). \end{aligned} \tag{3}$$

Taylor's formula for $(1 - p/m)^k$ proves this: the quadratic coefficient differs by $\lambda_N - p^2/(2m)$ and the cubic remainder is at most $p^3 k(k-1)(k-2)/(6m^2)$. For $k = 0, 1, 2$ the corresponding missing terms are zero.

The OU/Chebyshev multiplier bound and Gaussian power moments, followed by the ordinary tail of [2], imply for all fixed nonnegative a,b,j

$$\begin{aligned} \sup_{m, |s| \le S} \left\| h^a \partial_s^j (E_m - \mathsf{G}_m) h^b \right\|_{r,m} &= O(p^{-1}), \\ \sup_{m, |s| \le S} \left\| h^a \partial_s^j \mathsf{G}_m(s) h^b \right\|_{r,m} &< \infty. \end{aligned} \tag{4}$$

Indeed each coefficient has a polynomial-in-degree times $L_r^n \sqrt{n!}$ envelope; finitely many derivatives and exterior powers only change its constants and polynomial. Their exponential series are absolutely summable. Since $1 - \alpha = O(p^{-1})$,

$$\sup_{m, |s| \le S} \| \mathscr{Q}_\alpha[E_m] - \mathscr{Q}_1[\mathsf{G}_m] \|_{r,m} = O(p^{-1}). \tag{5}$$

No differentiation of an unweighted convergence statement is used.

3 Uniform total-memory norms before conditional regression

Set $U_s = e^{i\alpha hs}$ and define exact finite-N components

$$A_N(s) = (1-\mathcal{P}_h)e^{-ihs/2}\mathscr{Q}_\alpha[E_m](s)e^{-ihs/2},$$
$$B_N(s) = (1-\mathcal{P}_h)m\epsilon e^{-ihs/2}\Phi_m(U_s)e^{-ihs/2}.$$

Equation (5) bounds A_N in every fixed finite norm. The exact neutral series of [3] bounds B_N: $p(1-\mathcal{P}_h)\Phi_m(e^{ish}) = (1-\mathcal{P}_h)\sum_j c_{j,m}(s)T_{j,m}$, with summable actual coefficient bounds and $m\epsilon/p \le 2$.

The auxiliary average in (2) is even in $\sqrt{\epsilon}$. A fourth-order Duhamel remainder gives

$$\Lambda_{\mathrm{OU}}(s) = \epsilon\Phi_m(U_s) - \epsilon^2\mathcal{I}_s^{(\alpha)} + \mathcal{R}_s,$$
$$\mathcal{I}_s^{(\alpha)} = s^2\int_{0\le y\le x\le 1}\mathbb{E}_u[uU_{s(1-x)}uU_{s(x-y)}uU_{sy}u]\,dx\,dy, \tag{6}$$
$$\|\mathcal{R}_s\|_{r,m} \le \epsilon^3|s|^4\,\|u\|_{6r,m}^6/24. \tag{7}$$

The norm of u includes its independent Gaussian average. Conditional Jensen and Holder prove (7); all intermediate propagators are unitary and six u factors remain.

All three four-u pairings must be kept. For deterministic unitaries X,Y,Z and a real Gaussian coefficient basis $u = \sum g_iB_i$, $\sum_i B_i^2 = C_mI$, their sum is

$$\Phi(X)Y\Phi(Z) + \Phi(X\Phi(Y)Z) + \sum_{i,j}B_iXB_jYB_iZB_j.$$

Each term has operator norm at most C_m^2. For the last, use row–column Cauchy–Schwarz on B_iXB_j and YB_iZB_j; both square sums equal C_m^2I. Consequently

$$\left\|\mathcal{I}_s^{(\alpha)}\right\|_\infty \le \tfrac{3}{2}C_m^2s^2 \tag{8}$$

samplewise in the original h. This reproduces the existing auxiliary-field bound [7, 9] with an explicit factorization.

Let X_0 use Λ instead of M in $\mathcal{F}_N$. Since $m\epsilon^2 = O(1)$ and $m\epsilon^3 = O(p^{-1})$ on the central window, (6) implies

$$X_0 = A_N + B_N - (1-\mathcal{P}_h)m\epsilon^2e^{-ihs/2}\mathcal{I}_s^{(\alpha)}e^{-ihs/2} + O_{r,S}(p^{-1}).$$

The preceding return theorem [1] gives $\mathcal{F}_N - X_0 = O_{r,S}(p^{-1/2})$. All terms on the right are uniformly bounded. We have therefore proved

$$\sup_N\sup_{|m-N/2|\le L_0p}\sup_{|s|\le S}\|\mathcal{F}_N(s)\|_{r,m} < \infty. \tag{9}$$

This bound precedes the asymptotic removal of the four-u term.

4 Half-filled colored words and four-field regression

Now restrict to $m = N/2$. Let $H^{(0)}, H^{(1)}, H^{(2)}$ be independent coupling copies with the same CAR coefficient basis, and covariance $C_m \to C = J_s^2/\lambda$. Gaussian pairing permits only equal-color endpoints. For each allowed fixed pairing π, the independent p-set labels and occupation trace are identical to its uncolored counterpart.

Apply the central ordinary Fourier theorem [4] with all marks absent. Its integrand after $t = p\theta$ has limit $e^{-t^2/(8\lambda)} q^{\mathrm{cr}\,\pi}$, $q = e^{-4\lambda}$, and an integrable Gaussian majorant. The far Fourier interval is exponentially negligible. Exact half-charge extraction divides by the same unmarked Gaussian integral. Thus

$$\lim_N \mathbb{E}\,\mathrm{tr}_m \prod_{j=1}^{2k} H^{(c_j)} = C^k \sum_{\substack{\pi \text{ pairing} \\ c_i = c_j \text{ on } \pi}} q^{\mathrm{cr}\,\pi}. \tag{10}$$

Odd words vanish. This is a direct fixed-sector argument. It uses the whole Fourier domination, not restriction of a full-trace average. Only a finite set of pairings occurs at fixed degree.

Let h,u,v be orthogonal variance-C q-Gaussian variables in their tracial algebra. BKS second quantization, Theorems 2.11 and 2.13 [12], supplies trace-preserving conditional expectations E_h, E_{hv}. On Wick tensors, projection onto (h, v) removes u tensors exactly as projection onto h does. Therefore

$$\tau[W_1(h,u)^\dagger W_2(h,v)] = \tau[(E_h W_1)^\dagger (E_h W_2)]. \tag{11}$$

This is an identity for two whole blocks, not classical independence for arbitrary interleavings.

For fixed polynomials f_i put $W_N(h,u) = u f_1(h) u f_2(h) u f_3(h) u$ and $T_N(h) = \mathbb{E}_u W_N(h,u)$. Conditional independence gives the exact finite-N identity

$$\mathbb{E}_h\,\mathrm{tr}_m\, T_N^\dagger T_N = \mathbb{E}_{h,u,v}\,\mathrm{tr}_m\, W_N(h,u)^\dagger W_N(h,v).$$

Let the polynomial t satisfy $t(h) = E_h W(h,u)$. Applying (10) and (11) to this expression and its two mixed terms yields

$$\|T_N(h) - t(h)\|_{2,m} \longrightarrow 0. \tag{12}$$

The squared norm has limiting terms $\|t\|_2^2 - 2\,\|t\|_2^2 + \|t\|_2^2 = 0$. The conditional expectation is a known q-Gaussian construction; the finite-sector bridge here comes from (10).

5 Finite time and the complete two-component reduction

For $\alpha = 1$, grouping the ordinary energy powers by total degree gives the exact series

$$\mathcal{I}_s^{(1)} = \sum_{n\geq 0} \frac{s^2(is)^n}{(n+2)!} \sum_{a+b+c=n} \mathbb{E}_u[uh^a uh^b uh^c u]. \tag{13}$$

Gaussian moments and Holder bound its degree-n L^2 norm by

$$\frac{S^{n+2}}{2n!}\sqrt{(2n+7)!!}\, C_*^{(n+4)/2}.$$

This is summable for every fixed S, uniformly in N. It is the existing time-tail argument of [9], now using (12) in the half-filled sector.

Define $\iota_s(h)$ by replacing $\mathbb{E}_u$ in (13) by the q conditional expectation, or equivalently by its three bounded unitary factors. BKS's C^* result makes it a continuous function on $[-E_*, E_*]$, $E_* = 2\sqrt{C/(1-q)}$, with norm at most $S^2E_*^4/2$. Clip its energy argument to that interval to obtain a bounded continuous $\widetilde{\iota}_s$ on the real line. At each fixed degree, spectral weak convergence plus higher moments compares the polynomial target with this extension. Use (12), the uniform tail, and then a finite time mesh to obtain

$$\sup_{|s|\leq S} \left\| \mathcal{I}_s^{(1)} - \widetilde{\iota}_s(h) \right\|_{2,m} \to 0.$$

Both full-time terms have samplewise bounded operator norms, by (8) and the clipped target. Hence interpolation gives the same convergence for every fixed finite r.

Telescoping the three phases in $\mathcal{I}_s$ and applying Holder to the four u's and one h gives

$$\sup_{|s|\leq S} \left\| \mathcal{I}_s^{(\alpha)} - \mathcal{I}_s^{(1)} \right\|_{r,m} \leq c_{r,S}|1-\alpha| = O(p^{-1}). \tag{14}$$

The middle factors are unitary; $5r$ moments suffice. Because $\widetilde{\iota}_s(h)$ is exactly conserved, its phase-centered version is annihilated by $1 - \mathcal{P}_h$. Combining the result with (6) and the return theorem proves, at half filling,

$$\sup_{|s|\leq S} \|\mathcal{F}_N(s) - A_N(s) - B_N(s)\|_{r,m} \to 0. \tag{15}$$

No quantitative rate is asserted for the conditional-regression part. Energy pinching remains the finite-N samplewise projection, and no infinite time average is interchanged with N.

The centered four-u remainder was only bounded in (9); (15) now removes it strongly. This order separates the uniform-norm argument needed for actual propagation from the identification of the limiting field.

6 The common local memory field

Let $Q_d^{(q,C)}$ be monic q-Hermite polynomials and define $a_{nd}^{(q,C)}$ by $H^n = \sum_d a_{nd}^{(q,C)} Q_d^{(q,C)}(H)$. Use the physical-energy local charged fields J_d and neutral residuals N_j from [4, 3, 5]; all come from the original disorder. In terms of the dimensionless fields of [4], $J_d = C^{d/2} j_d$. Set

$$R_n = \sum_d a_{nd}^{(q,C)} J_d, \quad R_0 = R_1 = 0,$$
$$L_n = R_{n+2} - \{H, R_{n+1}\} + H R_n H. \tag{16}$$

The central ordinary same-copy bridge gives, at each fixed n, the actual unnormalized coefficient $[p\mathcal{N} - D - \lambda_N \mathcal{N}(\mathcal{N}-1)]H^n$ minus its deterministic energy polynomial equal to $R_{n,N} + o_{S_r}(1)$. It also controls fixed exterior energy powers. Applying $\mathscr{Q}_1$ preserves energy functions, so exact pinching removes those polynomials. Exact pinching of the residual local fields vanishes by their already proved two-energy measures and higher moments [4, 5, 3].

The ordinary tails, including two time derivatives, and the neutral actual/local tails are uniform at every fixed finite norm. Thus (5) and (15) give the limiting memory field

$$\mathcal{F}_\infty(s) = e^{-iHs/2} \left[\sum_{n\geq 0} \frac{(is)^n}{n!} L_n + 2C \sum_{j\geq 1} c_j^{(q,C)}(s) N_j \right] e^{-iHs/2}. \tag{17}$$

The factor $2C$ uses $m\epsilon/p \to 2$, only after a fixed coefficient truncation. Finite C_m, q_N, α are retained until that step. All series converge in every fixed finite norm inherited from the positive microscopic moments.

More precisely, there are same-sample local representatives $\mathcal{F}_N^{[J]}$ such that

$$\lim_{J\to\infty} \limsup_{N\to\infty} \sup_{|s|\leq S} \left\| \mathcal{F}_N(s) - \mathcal{F}_N^{[J]}(s) \right\|_{r,N/2} = 0. \tag{18}$$

The proof takes N to infinity at fixed J and then removes J using the factorial tails. It does not require an all-degree uniform error for the source asymptotics.

The joint law of the entire local tower is the central marked source law of [5, 4]. Neutral primitives have the form $\mathcal{W}_d[H^a((vp - L)Z_v \diamond Z_w)]$; the mark lies in a pure Wick subblock and exterior H slots may contract. These are within that theorem's scope. Its selected-boundary extension fixes the common moments with fixed preparations. One must retain this full tower; the low-degree H, size and curvature fields alone do not determine its joint moments.

7 Actual bilocal reconstruction at order one over p

Write $M_0 = \mathcal{P}_h M$, $\delta M = M - M_0$, and $\mathcal{Q}_N = m\delta M = e^{ihs/2}\mathcal{F}_N(s)e^{ihs/2}$. The exact Volterra equation is $F' = iaF - M * F$, $F(0) = I$, where convolution preserves order. Let F_0 solve the same equation with M_0.

We recall the explicit static dilation of [10], which is essential for its contraction. If $h = \sum_e eP_e$, take

$$D_0 = \bigoplus_e D_\perp, \quad R_0\psi = (RP_e\psi)_e, \quad K_0 = \begin{pmatrix} a & R_0^\dagger \\ R_0 & D_0 \end{pmatrix}.$$

The system compression of e^{itK_0} is F_0 because $R_0^\dagger e^{isD_0}R_0 = M_0(s)$. Therefore

$$\|F(t)\|_\infty, \|F_0(t)\|_\infty \le 1, \qquad [F_0(t), h] = 0.$$

Whole degenerate blocks are retained; F_0 need not be scalar inside them. The exact resolvent identity gives

$$F - F_0 = -F_0 * \delta M * F, \qquad m(F - F_0) = -F_0 * \mathcal{Q}_N * F. \tag{19}$$

The saved force moments [1] give $\|G\|_{r,m} = O(p^{-1})$ and $\|M(s)\|_{r,m}, \|M_0(s)\|_{r,m} \le \|G\|_{r,m}$. Let $U_h(t) = e^{iht}$ and $U_a(t) = e^{iat}$. Duhamel and the two contractions imply

$$\begin{aligned} \|F(t) - U_a(t)\|_{r,m}, \|F_0(t) - U_a(t)\|_{r,m} &\le \tfrac{1}{2}t^2 \|G\|_{r,m}, \\ \|U_a(t) - U_h(t)\|_{r,m} &\le |t|\frac{p}{m}\|h\|_{r,m}. \end{aligned} \tag{20}$$

Hence both propagator differences from U_h are $O(p^{-1})$ at every fixed finite r. By (9), $\|\mathcal{Q}_N(s)\|_{2r,m} = O(1)$. Subtract $-U_h * \mathcal{Q}_N * U_h$ from (19):

$$m(F - F_0) + U_h * \mathcal{Q}_N * U_h = -(F_0 - U_h) * \mathcal{Q}_N * F - U_h * \mathcal{Q}_N * (F - U_h).$$

Each integrand is $O(p^{-1})$ by $2r$ Holder and the remaining contraction. The simplex has bounded area. Centering and applying $1 - \mathcal{P}_h$ costs at most two, while the F_0 contribution is zero.

For left/right times v,w and middle time s with $v + w + s = t$, the centered integrand is $e^{ih(v-w)/2}\mathcal{F}_N(s)e^{-ih(v-w)/2}$. Put $u = (v - w)/2 = v - (t - s)/2$, so $du = dv$. Its pinching is already zero. This proves (1), including its negative sign and integration domain. It also gives a uniform norm bound on $\mathcal{B}_N$ by (19).

8 Selected covariant blocks and limiting bilocals

For $t \geq 0$ define the common limiting bilocal by

$$\mathcal{B}_\infty(t) = -\int_0^t ds \int_{-(t-s)/2}^{(t-s)/2} du\, e^{iHu} \mathcal{F}_\infty(s) e^{-iHu}, \qquad \mathcal{B}_\infty(-t) = \mathcal{B}_\infty(t)^\dagger. \tag{21}$$

The integral acts with norm at most $t^2/2$ on each fixed finite-norm field space. Equations (1) and (18) transfer its local truncations to the actual finite-N bilocal at half filling. All relevant products use the same disorder, preserving charged–neutral cross terms.

Let $P_N|O\rangle, Q_N|O\rangle$ be fixed polynomial selected preparations in the previously controlled class [6, 5]: their Gaussian degrees are fixed, their joint source moments exist, and their finite-N orbit densities obey uniform fixed moment bounds. For a covariant integrable Y,

$$\phi_N(P_N^\dagger Y P_N) = \mathbb{E}\,\mathrm{tr}_m(R_{P,N} Y), \qquad \sup_N \|R_{P,N}\|_{2,m} < \infty.$$

Polarization gives the analogous off-diagonal functional for P_N, Q_N. Thus an S_2 error in a whole covariant block has vanishing matrix element between these preparations.

For any fixed finite number of bilocal factors, trace Holder and telescoping put their product error in S_2: choose a sufficiently high but fixed norm for each factor and use (18), (1) and uniform moments. This applies to arbitrary prescribed order and adjoints, including out-of-time order, with fixed or compact external center times T_i. At a fixed local truncation, Gaussian energy Taylor tails treat the phases. The established marked source theorem then identifies the joint prepared moments. Consequently

$$\begin{aligned} &\lim_N \phi_N \left(P_N^\dagger \prod_{i=1}^k [e^{ihT_i} \mathcal{B}_N(t_i) e^{-ihT_i}]^{\eta_i} Q_N \right) \\ &\qquad = \phi \left(P^\dagger \prod_{i=1}^k [e^{iHT_i} \mathcal{B}_\infty(t_i) e^{-iHT_i}]^{\eta_i} Q \right), \qquad \eta_i \in \{1, \dagger\}. \end{aligned} \tag{22}$$

These are unnormalized matrix elements; a normalized preparation requires its limiting squared norm to be nonzero.

The single-selected restriction on the full covariant tower retains the known size boundary term [5, 3]:

$$\sigma_1(Y) = [1 - \sqrt{q}(1+\lambda)]\tau(Y) + \sqrt{q}\,\tau((\widehat{H} - W)Y(\widehat{H} - W)), \quad \widehat{H} = H/\sqrt{C}, \quad W = (S_1 - \lambda H)/\sqrt{C}.$$

The charged-only energy density cannot replace this expression for general neutral-containing bilocal words. Statement (22) covers one whole covariant block between fixed polynomial preparations. Arbitrary selected M factors interlaced inside several nonpolynomial blocks need further domain control. No bulk spatial/mass or finite-G gravitational dictionary is identified by this convergence theorem.

9 Finite evidence and remaining work

The fresh checks complement the analytic proof and inherited source/tail results; they are not double-scaling fits.

Interface	Checks
Unmarked central-source pairing constants	124
Three-color words: pairings versus q-Fock	1640
Whole-block auxiliary replica identities	400
Exact rational Gaussian-number remainders	105
Finite CAR charged/OU coefficient rows	10
Actual bilocal time points	6
Finite Schatten reconstruction inequalities	18

The color tests use $q = 1/4, 2/3$, variance one and exact rational arithmetic. The source constants cover all pairings through eight slots. Three-color words have up to six slots; the replica tests use four-auxiliary-field blocks and zero or one exterior energy gap. The number tests include chaos degree 80 and charge offsets $-p, 0, p$, checking the global multiplier bound rather than only low chaos.

CAR OU tests use $(N, p, m) = (3, 1, 2), (4, 1, 2), (4, 2, 3)$, seeds 811–813, $C_m = 1$, and all Gaussian coordinates shared between the adjacent charges. Direct conditional auxiliary Hermite averaging is compared with $\mathscr{Q}_\alpha$.

The propagation checks include a CAR fixture $(4, 2, 3)$ and an independent three-dimensional system with exact energy multiplicities $(2, 1)$ and a four-dimensional bath. Both have nonzero centered bilocal signals. The latter retains a non-scalar static-memory matrix inside the degenerate block, checking that pinching does not diagonalize within it. Times are $-0.55, 0.35, 0.9$; quadrature orders 10 and 18 are compared. Exact finite Volterra residuals, contractions, commutators, approximation differences and r=1,2,4 bounds are saved. Float64 comparisons use empirical roundoff tolerances, not interval certification.

The new rate in (1) uses the all-fixed-r bound on actual $m\delta M$; the older full-trace reconstruction used only a uniform second moment and a coarser higher-moment estimate [10]. Neither numerical refinement nor small finite systems prove the improved asymptotic rate.

The completed scope is the actual fixed-charge bilocal and its finite covariant blocks in controlled polynomial preparations. Next, its joint spectral/positive structure must be connected to a concrete static-observer interpretation, spatial and mass generators, and finite-G dynamics. Arbitrary nonpolynomial selected interlacings remain a separate mathematical interface. The build, input hashes, saved scientific output and rendered pages are recorded in this stage's local evidence manifest.

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